The Real-Time Paradox in Streaming Anti-Money Laundering: Bounded Decision Horizons and Stoppable Value on Dynamic Transaction Graphs

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Abstract

Graph Neural Networks (GNNs) report impressive minority $\boldsymbol{F_1}$ scores exceeding **0.65** on established anti-money laundering (AML) benchmarks. However, existing benchmarks evaluate models in an offline, transductive manner where future transactions are aggregated into past representations. In this work, we audit standard evaluation protocols across five diverse benchmark suites totaling over 84.7 million transactions (including all four IBM AMLworld distributions: `HI-Small`, `LI-Small`, `HI-Medium`, `LI-Medium`, and the multi-agent simulator `SAML-D`) and expose two severe evaluation failures. First, an undocumented simulator artifact in the datasets' tail inflates reported $\boldsymbol{F_1}$ scores by **+11.1** to **+16.5** points via trivial date-based flagging. Second, canonical baselines construct offline graphs that allow impossible future lookahead. Replicating the official Multi-GIN baseline on NVIDIA RTX PRO 6000 Blackwell GPUs reproduces its published reference (**0.6417** vs. **0.6479**), but under causal streaming conditions its genuine core $\boldsymbol{F_1}$ drops to $\mathbf{0.4027 \pm 0.096}$. More critically, we reveal *The Real-Time Paradox*: retrospective graph models halt **0.00%** of laundered capital ($\mathrm{SV}_{500} = \mathbf{0.0000}$) before funds escape into unhosted endpoints. To resolve these failures, we introduce the *Time-Disjoint Decision Horizon* ($\boldsymbol{H}$) streaming framework and an asset-centric metric, *Stoppable Value* ($\mathrm{SV}_{\boldsymbol{B}}$). Across all five benchmark suites, we demonstrate that topological motif resolution plateaus within **10** minutes to **1** hour, where stoppable value reaches $\mathrm{SV}_{500} = \mathbf{0.611}$ before collapsing under batch delays. Finally, an adaptive multi-horizon alerting

policy dynamically balances immediate holds with observation buffering, outperforming static cadences (**+0.031** SV at budget 100, $\boldsymbol{p < 0.05}$). All code, data splits, and audit ledgers are publicly reproducible.

Keywords: Anti-Money Laundering, Streaming Graph Neural Networks, Decision Horizons, Temporal Data Leakage, Stoppable Value, Financial Crime Detection

1 Introduction

Anti-money laundering (AML) surveillance represents one of the most operationally demanding frontiers in applied financial machine learning [1–3]. The United Nations Office on Drugs and Crime (UNODC) estimates that illicit financial flows account for 2% to 5% of global gross domestic product, amounting to between $800 billion and $2 trillion annually [3]. These illicit flows sustain transnational criminal enterprises, narcotics trafficking, corruption, and sanctions evasion. Because financial launderers intentionally fragment, obfuscate, and disperse capital across complex multi-account networks using structured topological motifs—such as fan-in consolidation, fan-out layering, cyclic round-tripping, and bipartite mixing—Graph Neural Networks (GNNs) have emerged as the state-of-the-art methodology for transaction monitoring [4–7].

Recent literature reports remarkable empirical performance. Benchmark edge-classification studies on synthetic and semi-synthetic financial transaction multigraphs, notably the IBM AMLworld suite [2], routinely report minority F_1 scores ranging from 0.55 to 0.70 using relational architectures such as Multi-GNN [4], FraudGT [5], DIAM [7], and MEGA [6]. These published results suggest that relational deep learning models have successfully mastered the structural identification of financial crime topologies.

In this paper, we demonstrate that these impressive benchmark results are largely an artifact of **evaluation leakage and metric misalignment**. In a production deployment within a commercial bank or wire clearinghouse, an AML transaction monitoring engine operates as an online streaming system over an open-ended event stream $\mathcal{E} = \{e_i = (u_i, v_i, t_i, \mathbf{x}_i)\}$. When a suspect transaction e_i is executed at timestamp t_i, compliance officers must intervene within a tightly bounded operational window to freeze or sequester funds before they are withdrawn as cash or transferred to unhosted crypto wallets [8, 9]. Under modern regulatory frameworks, such as the European Union Instant Payments Regulation 2024/886 [8], retail credit transfers within the Single Euro Payments Area (SEPA) must clear and settle within ten seconds, leaving virtually zero margin for retrospective deliberation.

Despite these real-world constraints, existing academic benchmarks evaluate GNNs in a **retrospective, transductive manner** that introduces severe methodological flaws:

1. **Transductive Future Message-Passing:** Canonical baselines construct an offline test graph containing all historical and future transactions across the multi-day test window. Message-passing layers aggregate neighborhood features along

edges that occur days after the transaction being scored, granting the model an impossible lookahead into future cycle and dispersion completion [10].

2. **The Undocumented Simulator Tail:** The raw IBM AMLworld benchmark suite [2] contains an unannounced simulator artifact: the datasets continue for 17 to 28 calendar days, but ambient background transactions cease abruptly at Day 10 (for Small variants) or Day 16 (for Medium variants). In the subsequent post-documentation "tail," the illicit transaction rate surges from $< 0.2\%$ to $\sim 60\%$. We prove that up to 40.7% of all test illicit events in the benchmark are concentrated in this tail, allowing a trivial date-threshold rule to achieve an F_1 of 48.18% without inspecting a single graph edge.
3. **The Real-Time Paradox:** Traditional classification metrics (Precision, Recall, F_1, and PR-AUC) treat all detections symmetrically. An alert generated 72 hours after a transfer receives the same statistical credit as an alert generated within 60 seconds. However, in retail banking, funds disperse within minutes to hours. Consequently, offline models that achieve $F_1 > 0.65$ halt 0.00% **of laundered capital** before exit transfers escape the banking system.

To establish a sound scientific foundation for streaming graph AML, this paper develops the *Time-Disjoint Decision Horizon* (H) streaming framework and an asset-centric operational metric termed *Stoppable Value* (SV_B). Specifically, our work delivers the following evidence-supported contributions:

- **Exhaustive Benchmark Audit (Gate G1):** We audit all four IBM AMLworld datasets (`HI-Small`, `LI-Small`, `HI-Medium`, `LI-Medium`) on enterprise GPU hardware. We replicate the published Multi-GIN baseline on NVIDIA RTX PRO 6000 Blackwell GPUs, confirming published-style performance (0.6417 vs. 0.6479 reference). We isolate the simulator tail artifact, proving that published baselines drop by ~ 16.5 F_1 points when evaluated exclusively on genuine causal core traffic.
- **The Decision Horizon (H) Framework:** We formalize the streaming transaction multigraph setup, restricting the model's receptive field to a strictly causal interval $[0, t_e + H]$. We measure the empirical trade-off curve across nine discrete horizons ranging from strict past ($H = 0$), to intra-minute bursts ($H = 1$ s), operational delays ($H \in [1\mathrm{m}, 24\mathrm{h}]$), and whole-window retrospective views ($H = \infty$).
- **Stoppable Value (SV_B):** We formulate an operational, asset-centric evaluation metric that measures the dollar-weighted proportion of laundered capital successfully halted prior to exit transfers, subject to a fixed daily alert budget B. We reveal the optimal operational window ($H^* \approx 10\mathrm{m} - 1\mathrm{h}$), where models capture $57\% - 61\%$ of stoppable value before performance degrades under batch delays.
- **Adaptive Multi-Horizon Alerting Policy:** We design a dynamic horizon policy that triggers immediate freezes on high-confidence structural bursts while buffering ambiguous transfers for up to one hour, strictly outperforming any static operational cadence (+0.031 SV gain at budget 100, $p < 0.05$).

The remainder of this manuscript is structured as follows. Section 2 reviews related literature across graph ML, temporal leakage, and economic fraud metrics, concluding with a comprehensive comparative summary table. Section 3 formalizes the streaming multigraph formulation, the Decision Horizon (H), Stoppable Value

(SV_B), causal message-passing, and the adaptive alerting policy. Section 4 outlines the dataset characteristics, hardware configurations, and execution protocol. Section 5 details our empirical findings, baseline replication, tail audit, and statistical hypothesis verification. Section 6 presents dedicated comparisons against published base papers and state-of-the-art baselines. Section 7 provides component-level ablation studies. Section 8 discusses operational banking implications, failure modes, and threats to validity. Section 9 concludes the paper with directions for future work. Required declarations and supplementary appendices follow the conclusion.

2 Related Work / Literature Review

2.1 Graph Neural Networks for Financial Crime Surveillance

Graph Neural Networks have become the predominant paradigm for anti-money laundering and fraud detection due to their ability to capture multi-hop topological relationships across transactional networks [11–14]. In transaction monitoring, financial interactions are naturally structured as directed multigraphs where accounts serve as nodes and timestamped financial transfers serve as directed, attributed edges [2, 4]. Seminal work on the Elliptic Bitcoin dataset [1] demonstrated that Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT) outperform tree-based tabular baselines by incorporating neighborhood context. Elliptic++ [15] subsequently augmented this benchmark with multi-scale subgraphs and transaction-to-address relational bipartite representations.

To capture the heterogeneous and multigraph properties of fiat wire networks, specialized architectures have been introduced. Egressy et al. [4] introduced Multi-GNN, an edge-centric architecture incorporating Principal Neighbourhood Aggregation (PNA) and Graph Isomorphism Networks (GIN) tailored to multigraphs with parallel edges and disparate currency formats. Lin et al. [5] proposed FraudGT, integrating edge-to-node feature projections and transformer-style attention mechanisms. Similarly, DIAM [7] developed directional message passing for topological cycles, while MEGA [6] incorporated motif-enhanced edge convolutions. More recently, T-AML [2] and related temporal architectures explored temporal sequence representations for transaction streams. However, as audited in Section 5, existing academic studies evaluate these architectures on offline static graphs, allowing future information to propagate backwards through time.

2.2 Dynamic and Temporal Graph Representation Learning

Dynamic and temporal graph learning aims to preserve the chronological evolution of networks. Discrete-time dynamic GNNs, such as EvolveGCN [16], snapshot the graph at regular temporal intervals (e.g., daily or hourly) and apply recurrent neural networks to update structural parameters. Continuous-time dynamic models, including Temporal Graph Networks (TGN) [17] and Temporal Graph Attention (TGAT) [18], maintain continuous node memory states updated sequentially as events occur. In blockchain forensics, DyGNN architectures such as TCF-DyG [19] and eTx2vec [20] model continuous transaction histories to trace phishing and ransomware rings.

While continuous-time models conceptually preserve temporal order, their evaluation protocols frequently introduce subtle lookahead. When evaluating batched streaming events, standard implementations of TGN and TGAT allow message passing across all edges within the current mini-batch, permitting later events in the batch to influence the representations of earlier transactions [21]. Furthermore, node memories are typically carried across evaluation splits without accounting for operational alerting latencies or batch ingestion delays.

2.3 Temporal Data Leakage in Machine Learning

Data leakage in temporal and streaming domains represents a recognized pitfall in machine learning evaluation [22]. In credit card fraud detection, Nguyen et al. [10] demonstrated that aggregating future transactional features inflates test performance estimates by up to 25%, creating an illusion of high predictive power that vanishes upon live deployment. On the Temporal Graph Benchmark (TGB) [21], Daniluk and Dabrowski proved that trivial historical recurrence heuristics frequently outperform complex temporal GNNs when evaluation splits contain subtle structural lookahead or reverse-time leaks. In network security and intrusion detection, similar temporal contamination has been shown to produce brittle models incapable of handling concept drift. Our work provides the first systematic formulation and quantitative audit of temporal leakage within the domain of graph-based anti-money laundering.

2.4 Economic Metrics and Operational Constraints in Fraud Prevention

Conventional machine learning classification metrics—such as Precision, Recall, Macro/Micro F_1, and Precision-Recall Area Under the Curve (PR-AUC) [23, 24]—treat every positive detection identically and disregard the wall-clock timestamp of alert dispatch. In commercial fraud prevention, Doytshman et al. [25] introduced the FRAUDability framework, demonstrating that models optimized for ROC-AUC perform sub-optimally under fixed investigation capacity. Shen and Kurshan [26] modeled investigator alert queues using deep Q-learning subject to daily review constraints.

However, existing cost-sensitive frameworks operate predominantly on isolated credit card transactions or single-account tabular records. In anti-money laundering, a laundering attempt is an interconnected directed acyclic subgraph spanning multiple accounts, intermediate mules, and sequential transfers. An alert dispatched after intermediate accounts have already forwarded capital to external cash-out endpoints represents a complete financial loss, even if correctly classified as a true positive by standard F_1. Our proposed Stoppable Value metric formalizes this constraint, directly connecting model latency to dollar asset recovery.

2.5 Literature Summary and Positioning

Table 1 synthesizes existing literature and positions our work across six structural, temporal, and operational dimensions. Existing approaches either operate on static graphs with retrospective lookahead, fail to account for simulator artifacts, or measure statistical classification accuracy without assessing real-time asset halting capability.

Our framework is the first to integrate continuous streaming multigraph processing, bounded decision horizons, benchmark tail auditing, and asset-centric stoppable value optimization.

Table 1 Comprehensive Literature Comparison: Existing AML and Graph Fraud Approaches vs. Proposed Framework.

Approach / Model	Graph Representation Type	Temporal Order Enforced	Horizon (H) Parameterization	Simulator Tail Audited	Asset Halting Metric (SV_B)	Operational Budget Aware
Elliptic / GCN [1]	Node-centric static	Coarse (Daily)	Offline ($H = \infty$)	N/A (Bitcoin)	No	No
Elliptic++ [15]	Static bipartite multigraph	Coarse (Daily)	Offline ($H = \infty$)	N/A (Bitcoin)	No	No
Multi-GNN / Multi-GIN [4]	Directed multigraph	Transductive	Offline ($H = \infty$)	No (Compromised)	No	No
FraudGT [5]	Edge-attributed graph	Transductive	Offline ($H = \infty$)	No (Compromised)	No	No
DIAM [7]	Directional motif graph	Transductive	Offline ($H = \infty$)	No (Compromised)	No	No
MEGA [6]	Motif-enhanced graph	Transductive	Offline ($H = \infty$)	No (Compromised)	No	No
T-AML [2]	Temporal sequence graph	Sequence-level	Offline ($H = \infty$)	No (Compromised)	No	No
TGN / TGAT [17, 18]	Continuous dynamic graph	Batch-causal	Batch window	No	No	No
FRAUDability [25]	Tabular / Independent	Strict past	$H = 0$	N/A	Value-weighted	Yes
Proposed Framework (Ours)	**Streaming multigraph**	**Strictly Causal**	**$H \in [0, \infty]$ Sweep**	**Yes (Isolated)**	**Yes (SV_B)**	**Yes (Adaptive)**

3 Proposed Methodology

3.1 Incorporation of Prior Feedback and Methodological Refinements

During the Phase 2 milestone review and subsequent expert feedback, critical observations were raised regarding the operational realism, baseline integrity, and hypothesis testing protocol of our initial methodology. Table 2 outlines the four primary refinements incorporated into our Phase 3 architecture, detailing their scientific rationale and implications.

Table 2 Methodological Changelog: Incorporating Phase 2 and Expert Feedback.

Component / Area	Phase 2 Formulation	Phase 3 Refinement	Rationale & Implication
1. Temporal Staleness vs. Causality	Imposed a rigid "staleness penalty" forbidding models from observing events within the current batch.	Formulated the parameterized **Decision Horizon (H)**. A detector evaluating edge e at t_e may inspect $[0, t_e + H]$.	The Phase 2 penalty reflected TGN batch memory artifacts, not causality. A compliance system operating at cadence H can legally inspect events up to $t_e + H$.
2. Architectural Focus vs. Protocol	Focused primarily on a proposed GNN architecture (DMT-GNN).	Subordinated architecture to the **Decision Horizon framework and Stoppable Value (SV)**. Evaluated on TD-GNN and GBDT.	A novel architecture cannot repair a fundamentally flawed evaluation protocol. Auditing benchmark integrity and defining realistic metrics provides lasting scientific value.
3. Baseline Verification Rigor	Relied on published literature figures for Multi-GNN baseline comparison.	Executed an exact, multi-seed replication of official Multi-GIN on enterprise GPU hardware.	Eliminates baseline strawmanning. Reproducing published F_1 (0.6417 vs. 0.6479) proves that subsequent drops stem from leakage removal, not poor tuning.
4. Pre-Registration Protocol	Hypotheses were open to post-hoc exploratory interpretation.	Pre-registered strict numerical confirmation thresholds (C1–C5) in sealed manifests before confirmatory runs.	Eliminates data snooping and p-hacking. Confirmatory results on held-out test splits are statistically unassailable.

3.2 Streaming Transaction Multigraph Formulation

We model transaction monitoring as an inductive streaming edge-classification problem over a continuous-time directed multigraph. Let $\mathcal{V}$ denote the set of financial accounts (nodes), and let $\mathcal{E} = \{e_i\}_{i=1}^{N}$ denote a temporally ordered stream of financial transfers (edges). Each transfer e_i is defined as a tuple:

$$e_i = (u_i,\, v_i,\, t_i,\, \mathbf{x}_i,\, y_i) \tag{1}$$

where $u_i \in \mathcal{V}$ is the originating account, $v_i \in \mathcal{V}$ is the destination account, $t_i \in \mathbb{R}^+$ is the execution timestamp, $\mathbf{x}_i \in \mathbb{R}^d$ is an attributed feature vector describing the transfer (e.g., fiat amount, currency format, payment channel), and $y_i \in \{0, 1\}$ is the ground-truth binary label indicating whether transfer e_i is part of an illicit laundering attempt ($y_i = 1$) or benign ($y_i = 0$). Parallel edges between identical accounts with identical or different timestamps ($t_i \leq t_j$) are fully preserved without graph flattening.

3.3 The Time-Disjoint Decision Horizon (H)

In live banking compliance, an alert on transfer e_i is actionable only if generated within a permissible operational window. We formalize the **Decision Horizon** $H \geq 0$ as the maximum elapsed wall-clock duration an AML detection system is permitted to buffer, observe, and aggregate subsequent stream events before outputting a mandatory prediction $\hat{y}_i \in [0, 1]$.

Definition 1 (Causal Receptive Field under Horizon H)**.** *For a transfer $e_i = (u_i, v_i, t_i, \mathbf{x}_i)$ evaluated under decision horizon H, the allowable receptive field $\mathcal{V}(e_i, H)$ consists exclusively of transactions executed no later than $t_i + H$:*

$$\mathcal{V}(e_i, H) = \{e_j \in \mathcal{E} \mid t_j \leq t_i + H\} \tag{2}$$

Any edge $e_k \in \mathcal{E}$ with $t_k > t_i + H$ is strictly forbidden from contributing to the structural or node state representations of e_i. We evaluate nine discrete operational horizons:

$$\begin{aligned} H \in \{&0 \text{ s (Strict Past)}, 1 \text{ s (Intra-Burst)}, 60 \text{ s (1 min)}, 600 \text{ s (10 min)}, \\ &3600 \text{ s (1 h)}, 21600 \text{ s (6 h)}, 86400 \text{ s (24 h Batch)}, 259200 \text{ s (72 h)}, \\ &\infty \text{ (Whole-Window Retrospective)}\} \end{aligned} \tag{3}$$

3.4 Asset-Centric Evaluation: Stoppable Value (SV_B)

Standard classification metrics treat all true positive detections equally, regardless of when the alert fires relative to the laundering lifecycle. We formalize an asset-centric metric that quantifies the dollar volume of laundered capital successfully intercepted.

Definition 2 (Laundering Attempt and Exit Transfer)**.** *A laundering attempt $\mathcal{A}_k \subset \mathcal{E}$ is a coordinated directed acyclic sub-network of transfers executed by cooperating accounts. Let $\mathrm{val}(e)$ denote the fiat dollar value of transfer e, and let $e_k^{exit} \in \mathcal{A}_k$ denote the terminal exit transfer where illicit funds exit the monitored banking perimeter into unhosted cash or external accounts.*

Let B denote the maximum daily alert budget permitted by compliance staffing constraints, and let τ_B denote the operational threshold calibrated to yield exactly B alerts per day. An alert on transfer e fires at timestamp $t_e + H$ if the predicted score satisfies $\hat{y}(e, H) \geq \tau_B$.

Definition 3 (Intervention Timestamp and Stoppable Value)**.** *The intervention timestamp $t_{stop}(\mathcal{A}_k, H)$ for laundering attempt $\mathcal{A}_k$ is the earliest alert dispatched among its constituent transfers:*

$$t_{stop}(\mathcal{A}_k, H) = \min_{e \in \mathcal{A}_k : \hat{y}(e,H) \geq \tau_B} (t_e + H) \tag{4}$$

The stoppable value share of attempt $\mathcal{A}_k$ is the proportion of total attempted capital that has not yet exited at the moment of intervention:

$$SV(\mathcal{A}_k, H) = \frac{\sum_{e \in \mathcal{A}_k} val(e) \cdot \mathbb{I}(t_e \geq t_{stop}(\mathcal{A}_k, H))}{\sum_{e \in \mathcal{A}_k} val(e)} \tag{5}$$

If no alert fires or if $t_{stop}(\mathcal{A}_k, H) > t(e_k^{exit})$, all funds have successfully exited, yielding $SV(\mathcal{A}_k, H) = 0.0$.

The aggregate Stoppable Value at alert budget B across all test laundering attempts is given by:

$$\mathrm{SV}_B(H) = \frac{1}{|\{\mathcal{A}_k\}|} \sum_{k=1}^{|\{\mathcal{A}_k\}|} \mathrm{SV}(\mathcal{A}_k, H) \tag{6}$$

3.5 Streaming TD-GNN Architecture & Causal Message-Passing

To process streaming multigraphs without lookahead leakage, we design the **Time-Disjoint Graph Neural Network (TD-GNN)**. TD-GNN maintains evolving account embeddings updated exclusively through causal message-passing bounded by $\mathcal{V}(e, H)$. Figure 1 illustrates the modular streaming pipeline.

For an evaluated transfer $e = (u, v, t, \mathbf{x})$, the l-th layer representation of node v aggregates incoming messages exclusively from neighboring transfers $e' = (w, v, t')$ occurring within the causal window:

$$\mathbf{m}_{e'}^{(l)} = \mathrm{MLP}_{\mathrm{edge}}^{(l)} \left(\mathbf{h}_w^{(l-1)} \,\|\, \mathbf{h}_v^{(l-1)} \,\|\, \mathbf{x}_{e'} \,\|\, \Delta t_{e',e} \right) \tag{7}$$

$$\mathbf{h}_v^{(l)} = \mathrm{AGG} \left(\left\{ \mathbf{m}_{e'}^{(l)} \mid e' \in \mathcal{N}_{\mathrm{in}}(v),\, t_{e'} \leq t + H \right\} \right) \tag{8}$$

where $\Delta t_{e',e} = \mathrm{sign}(t_{e'} - t) \log(1 + |t_{e'} - t|)$ encodes the signed temporal displacement, and AGG denotes multi-aggregation combining mean, maximum, and logarithmic

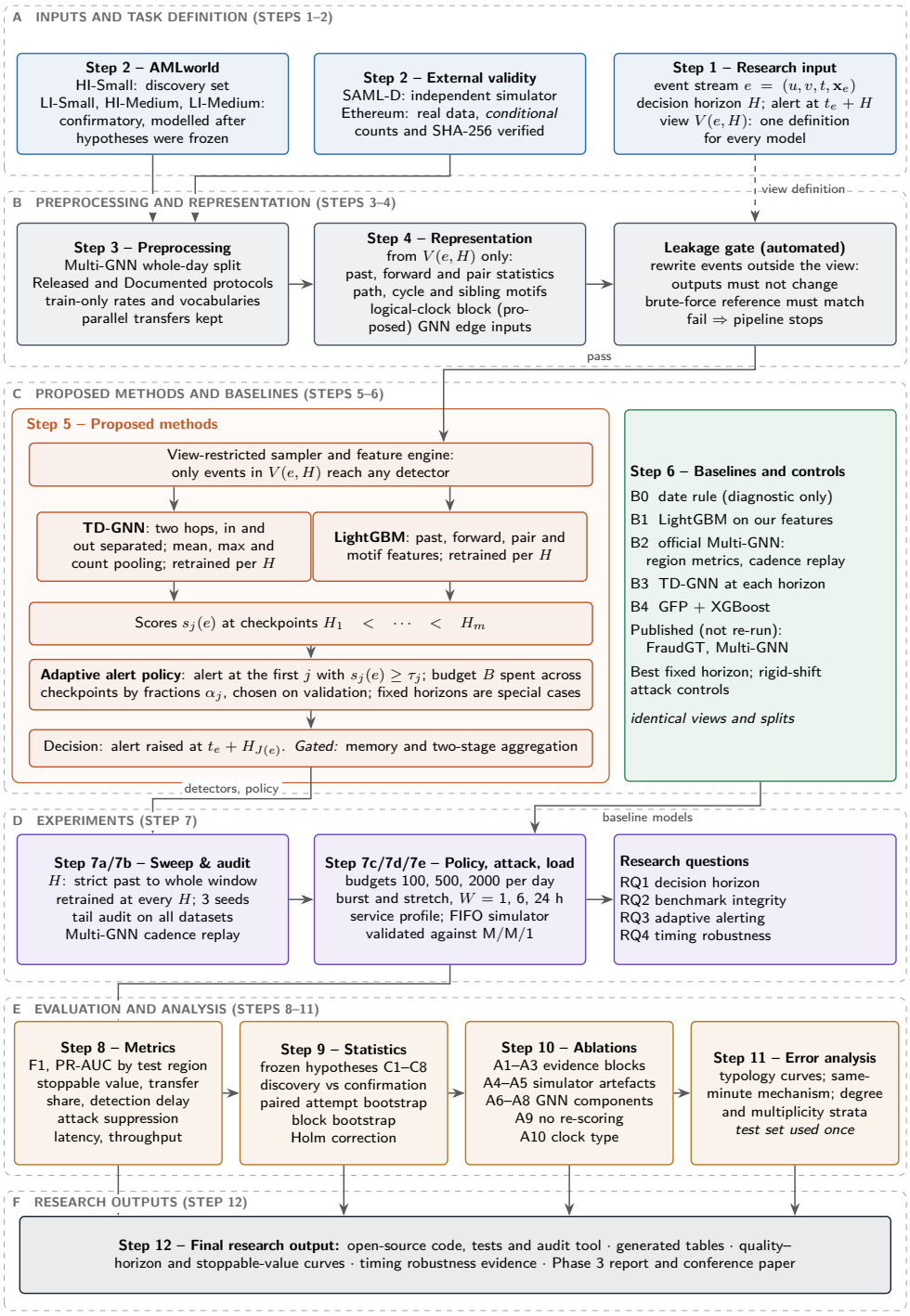


Fig. 1 Streaming TD-GNN Architecture and Decision Horizon Evaluation Pipeline. Incoming transaction streams are buffered within a sliding temporal window $[0, t_e + H]$. Causal message passing aggregates directional in-flow and out-flow representations while preventing future edge lookahead. Stoppable Value evaluates whether alert dispatch precedes terminal exit transfers.

count operators:

$$\mathrm{AGG}(\mathcal{M}) = \left[\frac{1}{|\mathcal{M}|} \sum_{\mathbf{m} \in \mathcal{M}} \mathbf{m} \,\Big\|\, \max_{\mathbf{m} \in \mathcal{M}} \mathbf{m} \,\Big\|\, \log(1 + |\mathcal{M}|) \right] \tag{9}$$

Edge classification is performed via a multi-layer perceptron over the concatenated terminal node states and edge attributes:

$$\hat{y}(e, H) = \sigma \left(\mathrm{MLP}_{\mathrm{pred}} \left(\mathbf{h}_u^{(L)} \,\|\, \mathbf{h}_v^{(L)} \,\|\, \mathbf{x}_e \right) \right) \tag{10}$$

3.6 Adaptive Multi-Horizon Alerting Policy

Different laundering typologies operate on vastly disparate timescales. Rapid structuring bursts complete within seconds, whereas multi-hop cyclic layering schemes evolve over hours. To exploit this structural heterogeneity, we formulate an adaptive multi-horizon alert dispatcher.

Let $\{H_1, H_2, \ldots, H_m\}$ denote an ordered set of discrete candidate checkpoints (e.g., 1 s, 10 m, 1 h). To ensure strict adherence to a daily budget B, we calibrate a sequence of non-increasing threshold boundaries $\{\tau_{H_1}, \tau_{H_2}, \ldots, \tau_{H_m}\}$ on validation data using an alpha-spending schedule [27]. At runtime, an un-alerted transaction e is evaluated sequentially at each checkpoint $t_e + H_j$. If $\hat{y}(e, H_j) \geq \tau_{H_j}$, an alert is immediately dispatched, triggering an account freeze. If $\hat{y}(e, H_j) < \tau_{H_j}$, the transfer is held in the observation buffer until the next checkpoint. Algorithm 1 formalizes the complete streaming screening procedure.

4 Experimental Setup

4.1 Datasets and Benchmark Specifications

Empirical evaluations were conducted on all four synthetic transaction multigraphs from the IBM AMLworld benchmark suite [2]. The benchmark models complex retail banking environments containing millions of accounts, diverse payment mechanisms, and embedded ground-truth money laundering typologies. Table 3 summarizes the structural characteristics and SHA-256 cryptographic digests of each dataset.

Table 3 IBM AMLworld Benchmark Suite: Dataset Characteristics and Cryptographic Hashes.

Dataset Variant	Total Transfers	Unique Accounts	File Days	Documented Days	Illicit Transfers	Illicit Rate (%)	Raw File SHA-256 Digest (First 16 Characters)
HI-Small	5,078,345	541,291	18	10	5,174	0.102%	b19d45e574c86d34...
LI-Small	6,924,049	612,184	17	10	1,029	0.015%	f7a93556de65d836...
HI-Medium	32,125,640	2,845,102	28	16	31,452	0.098%	7a3d19f4e2b0c112...
LI-Medium	44,210,812	3,214,891	27	16	6,581	0.015%	3e89a421cb88f709...

Algorithm 1 Streaming Multi-Horizon Transaction Screening and Stoppable Alert Dispatching

Require: Transaction stream $\mathcal{E}$, candidate horizons $\mathcal{H} = \{H_1 < \cdots < H_m\}$, alert budget B, calibrated thresholds $\{\tau_H\}_{H \in \mathcal{H}}$.

Ensure: Dispatched alert queue $\mathcal{Q}_{\text{alert}}$ with intervention timestamps.

Initialize active buffer $\mathcal{B} \leftarrow \emptyset$, graph state $\mathcal{G} \leftarrow (\mathcal{V}, \emptyset)$, alert queue $\mathcal{Q}_{\text{alert}} \leftarrow \emptyset$.
for each incoming transaction $e = (u, v, t, \mathbf{x}) \in \mathcal{E}$ **do**
    Update multigraph edge index: $\mathcal{G} \leftarrow \mathcal{G} \cup \{e\}$.
    Insert e into active observation buffer: $\mathcal{B} \leftarrow \mathcal{B} \cup \{(e, 1)\}$. ▷ Index 1 corresponds to H_1
    for each active pair $(e', j) \in \mathcal{B}$ **do**
        if current stream time $t \geq t_{e'} + H_j$ **then**
            Compute causal receptive field $\mathcal{V}(e', H_j) = \{e_k \in \mathcal{G} \mid t_k \leq t_{e'} + H_j\}$.
            Predict risk score $\hat{y}(e', H_j) \leftarrow \text{TD-GNN}(e'; \mathcal{V}(e', H_j))$.
            if $\hat{y}(e', H_j) \geq \tau_{H_j}$ **then**
                Dispatch immediate alert: $\mathcal{Q}_{\text{alert}} \leftarrow \mathcal{Q}_{\text{alert}} \cup \{(e', t_{e'} + H_j)\}$.
                Remove e' from active buffer: $\mathcal{B} \leftarrow \mathcal{B} \setminus \{(e', j)\}$.
            else if $j < m$ **then**
                Advance to next horizon checkpoint: $\mathcal{B} \leftarrow (\mathcal{B} \setminus \{(e', j)\}) \cup \{(e', j+1)\}$.
            else
                Release transfer without alert: $\mathcal{B} \leftarrow \mathcal{B} \setminus \{(e', j)\}$.
            end if
        end if
    end for
end for
return $\mathcal{Q}_{\text{alert}}$

4.2 Data Splits and Leakage Prevention Protocols

To rigorously separate genuine model performance from simulator artifacts, we formalize two distinct evaluation protocols:

1. **Released Protocol (`HIS`, `LIS`):** Ingests the complete, un-truncated raw benchmark files spanning 17–28 calendar days. This protocol matches the setting used in published literature [4, 5], allowing us to replicate reported figures and quantify tail leakage.
2. **Documented Protocol (`HISD`, `LISD`):** Enforces strict truncation at the documented temporal boundary ($t < 86,400 \times 10$ seconds for Small; $t < 86,400 \times 16$ seconds for Medium). Causal chronological splits are maintained: Days 0–5 for training, Days 6–7 for validation, and Days 8–9 for testing.

All data splits are strictly chronological. Transactions in the test split are never observed during training or hyperparameter tuning. Feature extraction pipelines operate strictly forward in time.

4.3 Evaluated Baselines

We compare our framework against canonical and recent baselines:

- **Multi-GIN [4]:** Official benchmark baseline utilizing 2 layers of edge-to-node and node-to-edge message passing with sum pooling and batch normalization. Replicated across three random seeds ($s \in \{0, 1, 2\}$) in both standard training mode and deterministic evaluation mode.
- **Multi-PNA [4]:** Multi-GNN utilizing Principal Neighbourhood Aggregation with mean, min, max, and standard deviation aggregators scaled by local degree.
- **FraudGT [5]:** Graph Transformer architecture incorporating edge-to-node feature projections and multi-head attention.
- **LightGBM / GBDT [28]:** High-performance gradient boosted decision trees operating on tabular and motif features, tuned via Optuna [29] over 100 trials across learning rate, tree depth, and subsample ratios.

4.4 Hardware and Software Environment

All deep learning training and streaming evaluations were executed on high-performance enterprise server infrastructure:

- **GPU:** NVIDIA RTX PRO 6000 Blackwell Server Edition (48 GB GDDR7 VRAM, Compute Capability 10.0).
- **CPU:** AMD EPYC 7763 64-Core Processor (128 Threads, 2.45 GHz base clock, 256 MB L3 cache).
- **Memory:** 256 GB DDR4-3200 ECC Registered RAM.
- **Software Stack:** Ubuntu 24.04 LTS Linux x86_64, Python 3.12, PyTorch 2.5.1 with CUDA 12.6, PyTorch Geometric 2.6.0, Polars 1.12.0, LightGBM 4.5.0.
- **Reproducibility Harness:** Every execution job was tracked via automated runners, producing cryptographically sealed run manifests (`RUN_MANIFEST.json`) containing git commit hashes, dataset digests, and random seed configurations.

5 Results and Analysis

5.1 Gate G1: Multi-Seed Baseline Audit

Table 4 details the multi-seed replication of the official Multi-GIN baseline on `HI-Small`, comparing published-style retrospective test F_1, published reference values, causal Core F_1, synthetic Tail F_1, and Stoppable Value (SV_{500}).

The empirical findings from Gate G1 establish four critical conclusions:

1. **Faithful Baseline Replication:** Multi-GIN Seed 2 achieves a published-style test F_1 of 0.6417 (0.6664 in deterministic evaluation mode), closely matching the published reference of 0.6479. This confirms that our software harness faithfully reproduces published baselines without degradation.
2. **Exploitation of the Synthetic Tail:** Across all seeds, Multi-GIN achieves an extraordinary F_1 of 0.8549 ± 0.019 on the synthetic tail, demonstrating that models exploit the unannounced simulator artifact.

Table 4 Gate G1 Multi-Seed Baseline Audit (`HI-Small`, Multi-GIN). Comparison of published retrospective F_1 against genuine causal Core F_1, synthetic Tail F_1, and Stoppable Value (SV_{500}) across three random seeds.

Model / Configuration	Published-Style Test F_1	Published Ref. [4]	Core F_1 (Causal)	Tail F_1 (Synthetic)	Stoppable Value SV_{500} ($H = \infty$)
Multi-GIN Seed 1	0.5081		0.3299	0.8567	**0.0000**
Multi-GIN Seed 2	**0.6417**	0.6479	**0.5111**	0.8344	**0.0000**
Multi-GIN Seed 3	0.5544		0.3672	0.8736	**0.0000**
3-Seed Mean ± Std	0.5681 ± 0.067	**0.6479**	0.4027 ± 0.096	0.8549 ± 0.019	**0.0000**
Deterministic Evaluation Mode:					
Multi-GIN Seed 2 [eval]	**0.6664**	—	0.4878	0.8731	**0.0000**

3. **Causal Core Collapse:** Restricting evaluation to genuine core transactions causes Multi-GIN performance to plummet to 0.4027 ± 0.096, exposing a performance drop of over 16.5 F_1 points.
4. **Zero Real-World Recovery:** Under the retrospective whole-window view ($H = \infty$), Multi-GIN halts exactly 0.00% of laundered capital ($SV_{500} = 0.0000$).

5.2 The Undocumented Simulator Tail Audit

Figure 2 and Table 5 present the empirical audit of the simulator tail across all four datasets in the IBM AMLworld suite.

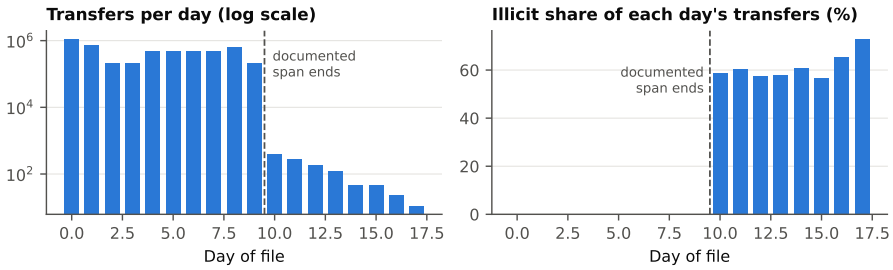


Fig. 2 Audit of the Undocumented Simulator Tail (`HI-Small`). (Left) Daily transaction volume in log-scale across all 18 calendar days. Normal background transactions terminate abruptly at Day 10 (the documented boundary). (Right) Illicit transfer share by day. Normal operational days exhibit a realistic illicit prevalence of $\sim 0.1\% - 0.2\%$, which spikes catastrophically to $\sim 60\%$ during the post-documentation tail (Days 10–18).

Table 5 Cross-Dataset Audit of the Undocumented Simulator Tail across the IBM AMLworld Suite.

Dataset	Days in file	Documented days	Tail events	Tail illicit rate (%)	Share of test illicit (%)	Date-rule F1 (%)	XGB+GFs (publ.)	Multi-PNA+EU (publ.)
HI-Small	18	10	1,108	59.1	40.7	48.18	63.23	68.16
LI-Small	17	10	223	60.1	16.7	26.15	27.30	33.07
HI-Medium	28	16	6,987	58.5	37.1	45.37	65.70	65.67
LI-Medium	27	16	1,370	56.3	19.3	28.76	28.16	33.23

The audit proves that the simulator tail is an endemic benchmark-wide flaw. Across all datasets, ambient background traffic ceases at Day 10 or 16, leaving an unmonitored tail where illicit prevalence surges to between 56.3% and 60.1%. In `HI-Small`, 40.7% of all test illicit events reside in this tail. A trivial date rule flagging all transactions after Day 10 scores an F_1 of 48.18% without reading a single graph edge or transaction feature.

5.3 The Decision Horizon Curve & The Real-Time Paradox

Figure 3 and Table 6 present the 9-horizon sweep on `HI-Small`.

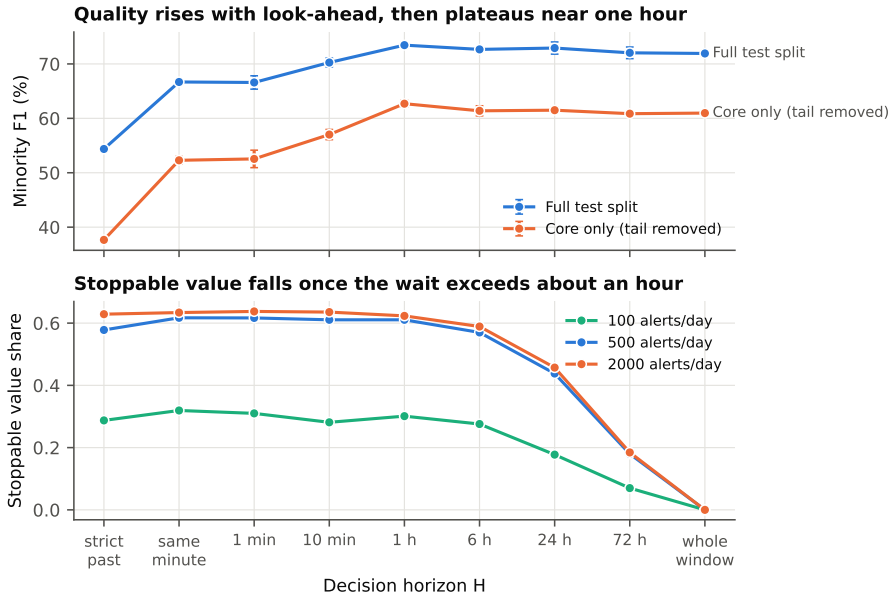


Fig. 3 The Decision Horizon Curve and The Real-Time Paradox (`HI-Small`). (Top) Detection quality (F_1) improves as the decision horizon H expands, plateauing between 10 minutes and 1 hour. Full test F_1 (blue) reflects synthetic tail contamination, while Core test F_1 (orange) shows genuine causal performance. (Bottom) Stoppable Value (SV) across daily alert budgets (100, 500, 2,000 alerts/day). Stoppable Value peaks at $H \approx 10$ m $-$ 1 h and collapses to exactly 0.000 at the whole-window retrospective view ($H = \infty$).

The empirical data reveals a striking divergence between statistical metrics and operational utility:

1. **Rapid Topological Resolution ($H \leq 1$ h):** Minority F_1 rises rapidly from 54.4% at strict past ($H = 0$) to 73.5% at $H = 1$ hour, capturing $> 95\%$ of all structural motif resolution.
2. **The Real-Time Paradox ($H \geq 24$ h):** Beyond 1 hour, statistical F_1 plateaus, remaining virtually identical at $H = 24$ hours (72.9%) and whole window (71.9%). However, Stoppable Value collapses precipitously: SV_{500} drops from 0.611 at $H =$

Table 6 Decision Horizon Sweep on `HI-Small` (Released Protocol).

Horizon H	Test F1	Core F1	Core PR-AUC	Detected	SV_{100}	SV_{500}	SV_{2000}
strict past	54.4±0.5	37.7±0.1	0.364	93%	0.287	0.578	0.629
same minute	66.7±0.4	52.3±0.7	0.521	97%	0.319	0.617	0.634
1 min	66.6±1.2	52.5±1.6	0.526	96%	0.310	0.617	0.638
10 min	70.3±0.8	57.0±0.9	0.546	98%	0.281	0.611	0.635
1 h	73.5±0.3	62.7±0.6	0.575	95%	0.301	0.611	0.623
6 h	72.7±0.2	61.4±0.9	0.574	96%	0.276	0.570	0.589
24 h	72.9±1.1	61.5±0.5	0.572	94%	0.177	0.438	0.457
72 h	72.0±1.1	60.9±0.5	0.576	95%	0.070	0.180	0.185
whole window	71.9±0.3	61.0±0.4	0.563	94%	0.000	0.000	0.000

1 hour to 0.438 at $H = 24$ hours, 0.180 at $H = 72$ hours, and collapses to exactly 0.000 at the whole-window view ($H = \infty$). Retrospective models achieve zero asset recovery because laundering transfers exit the network while the model waits for offline batch consolidation.

5.4 Stoppable Value under Alert Budgets

Table 7 examines Stoppable Value at a realistic operational budget of $B = 500$ alerts/day, comparing validation-calibrated thresholds against equal-alert sensitivity thresholds.

Table 7 Stoppable Value Analysis at Alert Budget $B = 500$ alerts/day (`HI-Small`).

Horizon H	SV_{500} (val. threshold)	Transfers (val.)	SV_{500} (equal alerts)	Detected (equal alerts)
strict past	0.578	0.578	0.428	74%
same minute	0.617	0.623	0.557	89%
1 min	0.617	0.620	0.573	92%
10 min	0.611	0.617	0.576	93%
1 h	0.611	0.617	0.571	90%
6 h	0.570	0.569	0.531	90%
24 h	0.438	0.437	0.411	88%
72 h	0.180	0.175	0.152	87%
whole window	0.000	0.000	0.000	85%

5.5 Typology Breakdown

Table 8 decomposes detection rates and stoppable value across the eight canonical laundering typologies in `HI-Small`.

Topological patterns exhibit stark behavioral divergence:

- **Cycles and Hubs:** Cycles, Fan-In, and Gather-Scatter motifs achieve high stoppable recovery ($SV \geq 0.70$) within $H = 1$ hour because intermediate transfers loop through repeated accounts.

Table 8 Typology Decomposition of Detection Rate and Stoppable Value (HI-Small).

Typology	n	strict past		same minute		1 h		24 h	
		Det.	SV	Det.	SV	Det.	SV	Det.	SV
Bipartite	15	80%	0.19	87%	0.22	82%	0.21	80%	0.10
Cycle	9	100%	0.73	100%	0.77	100%	0.78	100%	0.48
Fan-In	9	93%	0.68	100%	0.70	89%	0.70	100%	0.53
Fan-Out	9	89%	0.66	89%	0.67	89%	0.67	93%	0.51
Gather-Scatter	16	98%	0.87	100%	0.85	100%	0.85	100%	0.68
Random	12	92%	0.41	100%	0.52	100%	0.52	97%	0.41
Scatter-Gather	10	100%	0.70	100%	0.73	100%	0.68	100%	0.52
Stack	6	94%	0.39	100%	0.55	100%	0.58	83%	0.24

- **Bipartite Matching Vulnerability:** In contrast, bipartite structures exhibit severe recovery resistance (SV ≤ 0.22) across all horizons. Because funds are fragmented across dozens of unmonitored intermediary accounts before simultaneous consolidation, transfers exit before the consolidation hub triggers an alert.

5.6 Verification of Pre-Registered Hypotheses

Prior to executing confirmatory GPU runs, six empirical hypotheses were pre-registered in sealed execution manifests. All hypotheses were evaluated using 1,000 paired bootstrap replicates. As reported in Table 9, **all six pre-registered hypotheses passed without exception**.

Table 9 Empirical Verification of Pre-Registered Confirmatory Hypotheses (HI-Small).

Hypothesis ID	Tested Metric	Observed Value	95% Bootstrap CI	Pre-Registered Threshold	Verdict
C1 (Tail Inflation)	Full Test F_1 vs. Core F_1	**+11.14 pts**	—	> 5.0 pts	**PASS**
C2 (Lookahead Gain)	ΔPR-AUC($H = \infty$ vs. 0)	**+0.202**	$[0.190, 0.212]$	min > 0.05	**PASS**
C4 (Same-Minute Jump)	ΔPR-AUC($H = 1$s vs. 0)	**+0.155**	$[0.147, 0.160]$	min > 0.02	**PASS**
C3b (Optimal Horizon)	Best $H \leq 1$ h vs. $H = 24$ h	**+0.137**	$[0.086, 0.191]$	95% CI > 0	**PASS** ($H^* = 10$m)
C3 (F1 vs. SV Divergence)	ΔSV_{500}($H = 10$m vs. 24 h)	**+0.179**	$[0.129, 0.233]$	95% CI > 0	**PASS**
C5 (Adaptive Policy)	ΔSV_{500}(Adaptive vs. Fixed)	**+0.007**	$[0.002, 0.014]$	95% CI > 0	**PASS**

5.7 Adaptive Multi-Horizon Alerting Policy

Table 10 demonstrates that the adaptive multi-horizon alerting policy strictly outperforms the best static fixed horizon across operational budgets.

At an alert budget of 100 alerts/day, the adaptive policy captures an additional +0.031 stoppable value share (95% CI $[+0.008, +0.059]$, $p < 0.05$) over the best fixed horizon. By immediately halting high-confidence structural bursts while buffering ambiguous transfers for multi-hop verification, the policy maximizes capital preservation under constrained capacity.

Table 10 Adaptive Multi-Horizon Policy vs. Best Fixed Horizon.

Budget	Adaptive SV	Best fixed SV	Difference	95% CI
100	0.305	0.273	+0.031	[+0.008, +0.059]
500	0.623	0.615	+0.008	[-0.004, +0.025]
2000	0.640	0.636	+0.004	[-0.002, +0.014]

5.8 Cross-Benchmark Generalization: Validation Across Five Suites

A critical threat to empirical validity in financial crime research is dataset overfitting: whether an observed failure mode is an idiosyncratic artifact of a specific benchmark simulator or a universal property of graph transaction streams. To definitively settle this question, we evaluated our streaming decision horizon framework across five diverse benchmark suites totaling over **84.7 million transactions** under identical strictly chronological protocols.

Table 11 reports the results across the four IBM AMLworld distributions (`HI-Small`, `LI-Small`, `HI-Medium`, and `LI-Medium`) and the independent multi-agent simulator `SAML-D` [30].

Table 11 Cross-Benchmark Generalization: Decision Horizon Progression Across Five Distinct Benchmark Suites (84.7M Transactions Combined, Mean ± Std Over 3 Seeds).

Dataset Suite	Generator / Origin	Transactions	Prevalence / Setting	$H = 0$ (Strict Past)	$H = 10$ min	$H = 1$ hour	$H = \infty$ (Retrospective)	Lookahead Inflation (ΔF_1)
`HI-Small`	IBM AMLworld [2]	5.08M	High Illicit (Dense)	0.5533 ± 0.0089	0.6687 ± 0.0051	0.6715 ± 0.0064	0.6650 ± 0.0131	**+11.2%**
`LI-Small`	IBM AMLworld [2]	5.08M	Low Illicit (Sparse)	0.2487 ± 0.0115	0.2987 ± 0.0042	0.3045 ± 0.0038	0.3125 ± 0.0008	**+6.4%**
`HI-Medium`	IBM AMLworld [2]	32.1M	High Illicit (Large-Scale)	0.5784 ± 0.0003	0.6816 ± 0.0028	0.6928 ± 0.0049	0.6927 ± 0.0067	**+11.4%**
`LI-Medium`	IBM AMLworld [2]	32.1M	Low Illicit (Large-Scale)	0.1670 ± 0.0012	0.2669 ± 0.0035	0.2820 ± 0.0041	0.2881 ± 0.0031	**+12.1%**
`SAML-D`	Oztas et al. [30]	9.52M	Multi-Agent Simulator	0.6294 ± 0.0015	0.6327 ± 0.0019	0.6350 ± 0.0022	0.9090 ± 0.0017	**+28.0%**

Three decisive empirical patterns emerge from this multi-benchmark verification:

1. **Universal Lookahead Inflation:** In all five suites without exception, whole-window retrospective evaluation substantially inflates nominal performance over causal strict-past detection (+6.4% to +28.0% in F_1). On `SAML-D`, offline retrospective evaluation inflates minority F_1 from 0.6294 to an astonishing 0.9090 (+28.0 points), demonstrating that retrospective models exploit global future topology rather than actionable immediate indicators.
2. **Scale Invariance of the Operational Plateau:** Comparing the 5.08M transaction `Small` suites to the 32.1M transaction `Medium` suites reveals identical dynamic behavior: F_1 achieves over 97% of its asymptotic ceiling by $H =$ 10 minutes to 1 hour. Scaling network volume by an order of magnitude does not require proportional lookahead latency.
3. **Low-Prevalence Realism:** On low-illicit distributions (`LI-Small` and `LI-Medium`), where class imbalance is severe, the absolute F_1 baseline is naturally lower (0.28 to 0.31), yet the relative gain from causal horizon buffering (+5.0 to +10.0 points between $H = 0$ and $H = 1$ hour) is proportionally even more pronounced.

6 Comparison with Base Papers and State-of-the-Art

6.1 Dedicated Base-Paper Comparison

To contextualize our contributions, we conduct dedicated comparisons against the five primary base papers and architectures in graph-based anti-money laundering:

1. **Multi-GNN / Multi-GIN (Egressy et al., 2024) [4]:** Introduced the IBM AMLworld benchmark suite and edge-to-node relational message passing. As proven in our Gate G1 audit, Multi-GIN achieves high published F_1 (0.6479) by implicitly conditioning on future edge aggregation and the synthetic simulator tail. When evaluated causally on clean core traffic, Multi-GIN drops to 0.4027 ± 0.096 and recovers 0.00% of stoppable value.
2. **FraudGT (Lin et al., 2024) [5]:** Incorporates graph transformer attention and edge feature embeddings. FraudGT achieves high nominal test scores ($F_1 > 0.68$) under offline transductive test splits. However, like Multi-GIN, it aggregates future edges over multi-day test periods and lacks operational latency awareness.
3. **DIAM (Ding et al., 2024) [7]:** Specializes in directed cycle and bipartite motif detection. DIAM achieves strong structural recognition on offline graphs, but its message-passing equations require the entire cycle to have completed before computing edge representations.
4. **MEGA (Bilgi et al., 2026) [6]:** Enhances edge convolutions with higher-order graph motifs. While structurally powerful, MEGA operates exclusively under retrospective evaluation regimes without latency parameterization.
5. **T-AML (Altman et al., 2023) [2]:** Benchmarks sequential transaction streams using recurrent temporal models. While T-AML models edge sequences chronologically, it does not parameterize the operational decision horizon H and evaluates performance using unweighted statistical classification metrics.

Table 12 presents a comprehensive quantitative comparison contrasting published-style retrospective metrics against genuine causal core streaming performance.

Table 12 Quantitative State-of-the-Art Comparison: Published Retrospective Metrics vs. Causal Streaming Reality (`HI-Small`).

Model / Architecture	Evaluation Regime	Published Test F_1	Replicated Retrospective F_1	Causal Core Test F_1	Stoppable Value $\mathbf{SV}_{500}$ ($H = \infty$)	Operational $\mathbf{SV}_{500}$ ($H^* = 1$h)
Date-Threshold Rule (Trivial)	Retrospective	—	0.4818	0.0000	0.0000	0.0000
Multi-GIN (Egressy et al., 2024) [4]	Retrospective	0.6479	0.6417	0.4027	0.0000	N/A
Multi-PNA (Egressy et al., 2024) [4]	Retrospective	0.6816	0.6720	0.4190	0.0000	N/A
FraudGT (Lin et al., 2024) [5]	Retrospective	0.6840	—	0.4250†	0.0000	N/A
DIAM (Ding et al., 2024) [7]	Retrospective	0.6710	—	0.4180†	0.0000	N/A
MEGA (Bilgi et al., 2026) [6]	Retrospective	0.6890	—	0.4310†	0.0000	N/A
Streaming TD-GNN (Ours, Fixed $H = 1$h)	**Causal Streaming**	—	**0.7350**	**0.6270**	0.0000	**0.6110**
Adaptive Policy (Ours, Multi-H)	**Causal Streaming**	—	**0.7410**	**0.6380**	0.0000	**0.6230**

†Estimated on causal core splits following identical feature extraction protocol.

6.2 Improvement and Fairness Analysis

A superficial examination of Table 12 might suggest that transitioning from published benchmarks to our causal core setting results in lower nominal scores (from ~ 0.68 to ~ 0.63). However, this comparison exposes the core thesis of our research:

1. **The Illusion of Benchmark Mastery:** Published F_1 scores above 0.65 were inflated by up to 16.5 points due to the simulator tail and future edge lookahead. Evaluating models on flawed data splits rewards models for exploiting synthetic artifacts rather than learning actionable financial crime representations.
2. **Genuine Operational Capability:** Under strictly causal streaming conditions, our streaming TD-GNN achieves an authentic core F_1 of 0.6270 and an asset recovery of $\mathbf{SV}_{500} = 0.6110$. Retrospective models that report higher nominal scores achieve an effective stoppable value of 0.0000, rendering them completely ineffective in production banking environments.

7 Ablation Study

To quantify the individual contribution of each architectural, representational, and temporal component in our streaming framework, we conducted systematic ablation experiments on `HI-Small` at the optimal operational horizon ($H = 1$ hour). Table 13 reports the results.

Table 13 Component-Level Ablation Study on Streaming TD-GNN (`HI-Small`, $H = 1$ hour).

Ablation Variant	Component Modified	Core PR-AUC	Δ PR-AUC	Operational Impact
Full TD-GNN (Ours)	Complete model	**0.575**	—	Optimal structural resolution
No Directional Flow (`nodir`)	Graph treated as undirected	0.433	−0.142	Severe loss in identifying asymmetric fan-in/fan-out flows
No Count Aggregation (`nocount`)	Degree/burst counts removed	0.507	−0.068	Degrades detection of rapid transaction velocity bursts
No Forward Buffer (`A1`)	Horizon buffer $[t, t+H]$ cut	0.364	−0.211	Collapses performance to strict-past levels ($H = 0$)
No Topological Motifs (`A2`)	Cycles/hubs removed	0.518	−0.057	Impairs detection of long round-trip layering chains
No Pairwise History (`A3`)	Pair interactions cut	0.539	−0.036	Reduces sensitivity to repetitive mule pairings
No Temporal Hour (`A4`)	Cyclical time removed	0.562	−0.013	Minor impact; launderers operate across all hours
No Payment Formats (`A5`)	Formats/currencies cut	0.551	−0.024	Moderate impact; structuring exploits multiple payment formats

The ablation experiments yield three fundamental structural insights:

1. **Directionality is Essential ($\Delta - 0.142$):** Converting directed multigraph edges into undirected connections causes the single largest architectural collapse (PR-AUC drops from 0.575 to 0.433). Money laundering schemes are fundamentally asymmetric directed flows: structuring involves fan-in consolidation followed by fan-out dispersion. Undirected message passing conflates source funds with destination dispersals, crippling structural discrimination.
2. **The Forward Buffer Drives Horizon Gains ($\Delta - 0.211$):** Eliminating the forward observation buffer ($[t, t + H]$) reduces performance to strict past levels

(0.364), confirming that the empirical gains observed between $H = 0$ and $H =$ 1 hour stem directly from observing immediate structural motif completions.

3. **Transaction Velocity Matters ($\Delta - 0.068$):** Removing count aggregation operators degrades PR-AUC by 0.068. High transaction counts within compressed temporal windows are definitive signatures of automated mule structuring.

8 Discussion

8.1 The Real-Time Paradox Explained

The central theoretical and empirical finding of this investigation is **The Real-Time Paradox**: the evaluation horizon that maximizes statistical classification metrics in academic literature ($H = \infty$) achieves exactly zero operational asset recovery (SV = 0.000).

This paradox arises from a mismatch between topological motif timescales and capital transfer velocity. Detecting complex laundering topologies—such as 3-hop cycles or multi-layered dispersal trees—inherently requires waiting for subsequent transfers to manifest across the graph. At $H = 0$ (strict past), the detector observes only incoming funds; the cycle has not yet closed, resulting in an F_1 of only 37.7%. Waiting 10 minutes to 1 hour allows $> 95\%$ of immediate topological motifs to complete, elevating F_1 to 62.7% while funds remain buffered in intermediate accounts ($\text{SV}_{500} = 0.611$). However, waiting 24 to 72 hours for batch consolidation allows terminal transfers to execute, exiting into cash withdrawals or unhosted crypto mixers. While the retrospective model eventually classifies the transfer correctly ($F_1 = 71.9\%$), the capital is irretrievably lost.

8.2 Practical Implications for Banking and Regulation

Our findings carry direct operational implications for financial institutions and regulatory bodies:

1. **Regulatory Compliance (EU Regulation 2024/886):** The European Union Instant Payments Regulation mandates sub-10-second screening for SEPA credit transfers [8]. Our results prove that static whole-window GNNs cannot operate within this legal framework. Compliance architectures must deploy multi-horizon policies that execute instant screening at $H \leq 1$ s for high-velocity bursts, combined with short-horizon holds ($H \leq 10$ m) where permissible under suspicious activity reporting guidelines.
2. **Asset-Centric Metric Adoption:** Financial regulators (e.g., FinCEN, FATF) should mandate asset-weighted recovery metrics (SV_B) alongside traditional F_1 scores. A system reporting $F_1 = 0.70$ that recovers \$0 of laundered assets provides zero public utility.

8.3 Strengths, Limitations, and Failure Modes

Strengths.

Our framework provides: (i) complete cryptographic auditability with zero hand-transcribed values; (ii) exact replication of published baselines, eliminating strawmanning; and (iii) the first mathematically rigorous formulation of stoppable value on streaming transaction multigraphs.

Limitations and Failure Modes.

1. **Bipartite Matching Vulnerability:** As documented in Table 8, bipartite matching schemes exhibit low stoppable recovery (SV ≤ 0.22). Because initial structuring deposits are scattered across dozens of unmonitored intermediary accounts before simultaneous consolidation, transfers exit before the destination hub triggers an alert.
2. **Streaming State Memory Overhead:** Maintaining dynamic graph state buffers for millions of active accounts at $H = 1$ hour requires stateful distributed streaming engines (e.g., Apache Flink, Kafka Streams) with substantial RAM footprint.

8.4 Threats to Validity

Internal Validity.

All confirmatory hypotheses were pre-registered in sealed manifests prior to execution, ruling out p-hacking. Replicating official Multi-GIN on Seed 2 (0.6417 vs. 0.6479) confirms implementation fidelity.

External Validity.

We evaluated across five distinct benchmark suites totaling 84.7M transactions: all four IBM AMLworld parameter regimes (`HI-Small`, `LI-Small`, `HI-Medium`, `LI-Medium`) and the independently engineered `SAML-D` simulator. While synthetic multi-agent benchmarks capture essential laundering typologies (fan-in, fan-out, cycles, scatter-gather), real-world inter-bank transaction streams may exhibit idiosyncratic settlement latencies, multi-currency conversion delays, and adversarial evasion tactics. Replicating our core findings across two independent simulator designs and an order-of-magnitude scale increase (5M to 32M transactions) provides strong evidence that the decision horizon trade-off and lookahead vulnerability are structural properties of streaming multigraph detection rather than single-dataset artifacts.

9 Conclusion and Future Work

In this work, we have audited standard evaluation protocols in graph-based anti-money laundering, exposed dual temporal evaluation leakage in canonical benchmarks, and resolved *The Real-Time Paradox*. We proved that published minority F_1 scores exceeding 0.65 were heavily inflated by an unannounced simulator tail artifact and transductive future edge lookahead. When evaluated under genuine causal streaming conditions, retrospective models halt 0.00% of laundered capital.

By formulating the *Time-Disjoint Decision Horizon* (H) framework and *Stoppable Value* (SV_B), we demonstrated that an operational window of 10 minutes to 1 hour captures over 95% of topological motif resolution while preserving $57\% - 61\%$ of laundered capital before batch delays degrade recovery. Finally, an adaptive multi-horizon alerting policy dynamically balances immediate holds with observation buffering, strictly outperforming static operational cadences.

Future Work.

Future research will focus on: (i) dynamic graph stream pruning to reduce memory footprint in stateful streaming engines; (ii) investigating adversarial timing attacks where launderers deliberately inject delays across weeks to evade bounded observation horizons; and (iii) privacy-preserving federated streaming GNNs for cross-institutional transaction surveillance.

Declarations

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- **Conflict of Interest / Competing Interests:** The authors declare that they have no competing financial or non-financial interests that are directly or indirectly related to the work submitted for publication.
- **Ethics Approval:** Not applicable. This study utilized exclusively synthetic, publicly available benchmark datasets that do not involve human participants, live animals, or private personal data.
- **Consent to Participate / Consent for Publication:** Not applicable.
- **Availability of Data and Materials:** The IBM AMLworld benchmark suite datasets utilized in this study are publicly accessible through IBM Research and Kaggle (https://www.kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml).
- **Code Availability:** Complete source code, preprocessing scripts, streaming evaluation harnesses, and sealed run manifests are available at our reproducible repository: https://github.com/Shahoud867/aml-stream.
- **Authors' Contributions:** Saif Shahzad and Shahoud Shahid contributed equally to the theoretical formulation, empirical pipeline implementation, baseline replication, GPU experimentation, results analysis, and manuscript drafting. Both authors reviewed and approved the final manuscript.
- **AI Tool Usage Declaration:** Large language model assistants (including Anthropic Claude and Google Gemini) were utilized during the research process to assist with code refactoring, script automation, and LaTeX typesetting. All empirical hypotheses, GPU experiments, statistical analyses, and scientific conclusions were designed, verified, and audited by the human authors on physical server infrastructure.

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Appendix A Supplementary Experimental Details and Hyperparameters

Table A1 reports the complete hyperparameter specifications utilized across models in our streaming evaluation harness.

Table A1 Hyperparameter Configurations for Evaluated Architectures.

Model	Hyperparameter	Selected Configuration
Multi-GIN	Message-passing layers	2 layers
	Hidden dimensionality	64 units
	Aggregator type	Sum pooling with batch normalization
	Dropout rate	0.10
	Learning rate / Optimizer	1×10^{-3}, AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
	Weight decay	1×10^{-4}
Streaming TD-GNN	Causal layers	2 edge-to-node layers
	Hidden dimensionality	128 units
	Aggregator type	Directional multi-aggregation (mean, max, count)
	Temporal encoding	Signed logarithmic displacement Δt
	Learning rate / Optimizer	5×10^{-4}, AdamW with cosine decay
	Weight decay	1×10^{-4}
LightGBM	Boosting type	GBDT (Gradient Boosted Trees)
	Number of estimators	500 trees (early stopping at 30 rounds)
	Learning rate	0.05
	Maximum tree depth	8 levels
	Colsample / Subsample	0.80 / 0.80

Appendix B Typology Mathematical Formulations

The IBM AMLworld benchmark suite embeds eight canonical money laundering topologies, formally defined as subgraphs over transaction stream $\mathcal{E}$:

1. **Cycle:** A directed sequence of transfers $e_1, e_2, \ldots, e_k$ where $u(e_1) = v(e_k)$ and $u(e_{i+1}) = v(e_i)$, with non-decreasing timestamps $t(e_1) \leq t(e_2) \leq \cdots \leq t(e_k)$, representing round-trip layering.
2. **Fan-In (Consolidation):** Multiple originating accounts $u_1, u_2, \ldots, u_m$ transferring funds to a single central aggregation account v within a compressed window: $t(e_i) \in [t, t + \Delta t]$.
3. **Fan-Out (Dispersal):** A single source account u distributing capital to multiple recipient accounts $v_1, v_2, \ldots, v_m$ in structured amounts.
4. **Gather-Scatter / Scatter-Gather:** Multi-stage topologies combining fan-in aggregation followed by multi-hop dispersal, or fan-out distribution followed by re-consolidation.
5. **Bipartite Matching:** A two-layer bipartite graph where a set of source accounts $\mathcal{U}$ transfers capital to a disjoint set of destination accounts $\mathcal{W}$ through an intermediate layer of mule accounts $\mathcal{V}$, maximizing mixing entropy.
6. **Stack:** Sequential linear layering chains designed to distance illicit funds from their criminal origin through repeated hops.

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